

W1 · 操作手順

開發環境建置

Python 與 VS Code

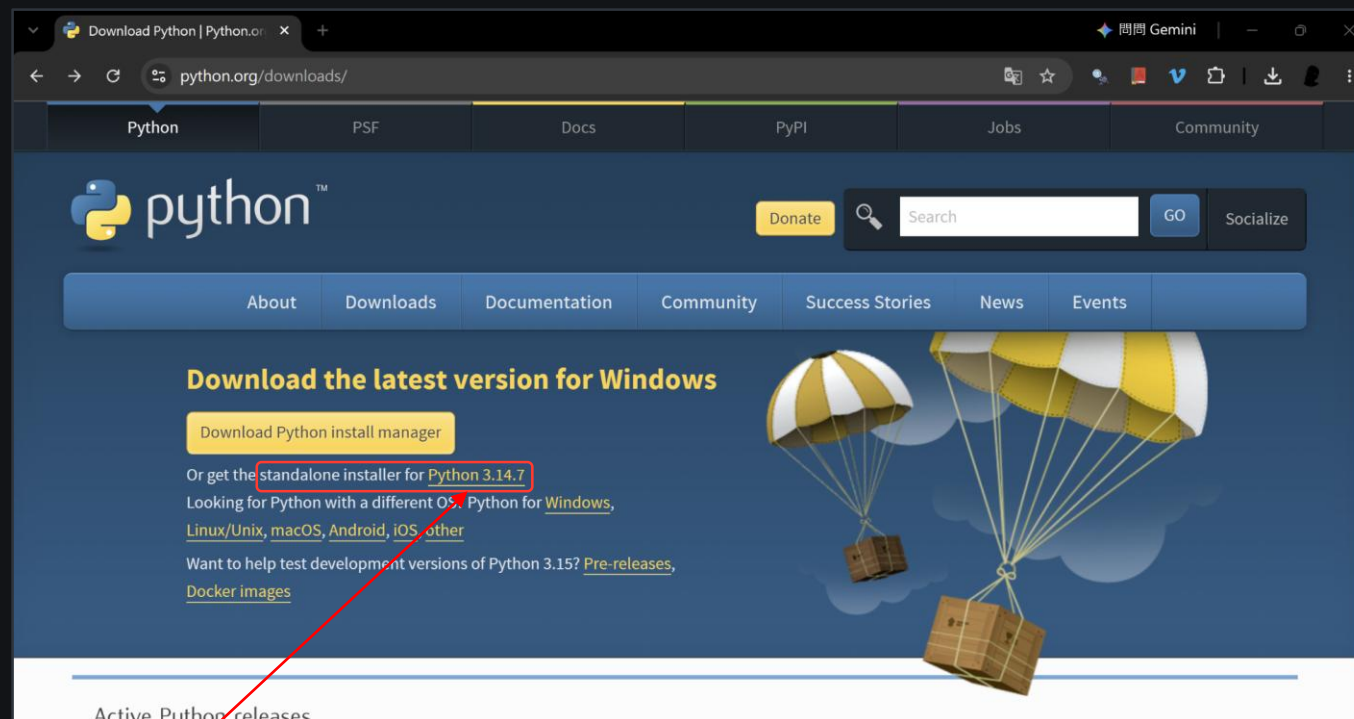
吳彥霖 博士

Python 網路爬蟲實務應用

步驟 1

下載且安裝 Python

官網：<https://www.python.org/downloads/>



安裝重點：

若是點選頁面「standalone installer」下載安裝檔；安裝時請勾選 **Add python.exe to PATH**，另外，安裝位置可自選（例如 D 槽）。

安裝時勾選 Add python.exe to PATH

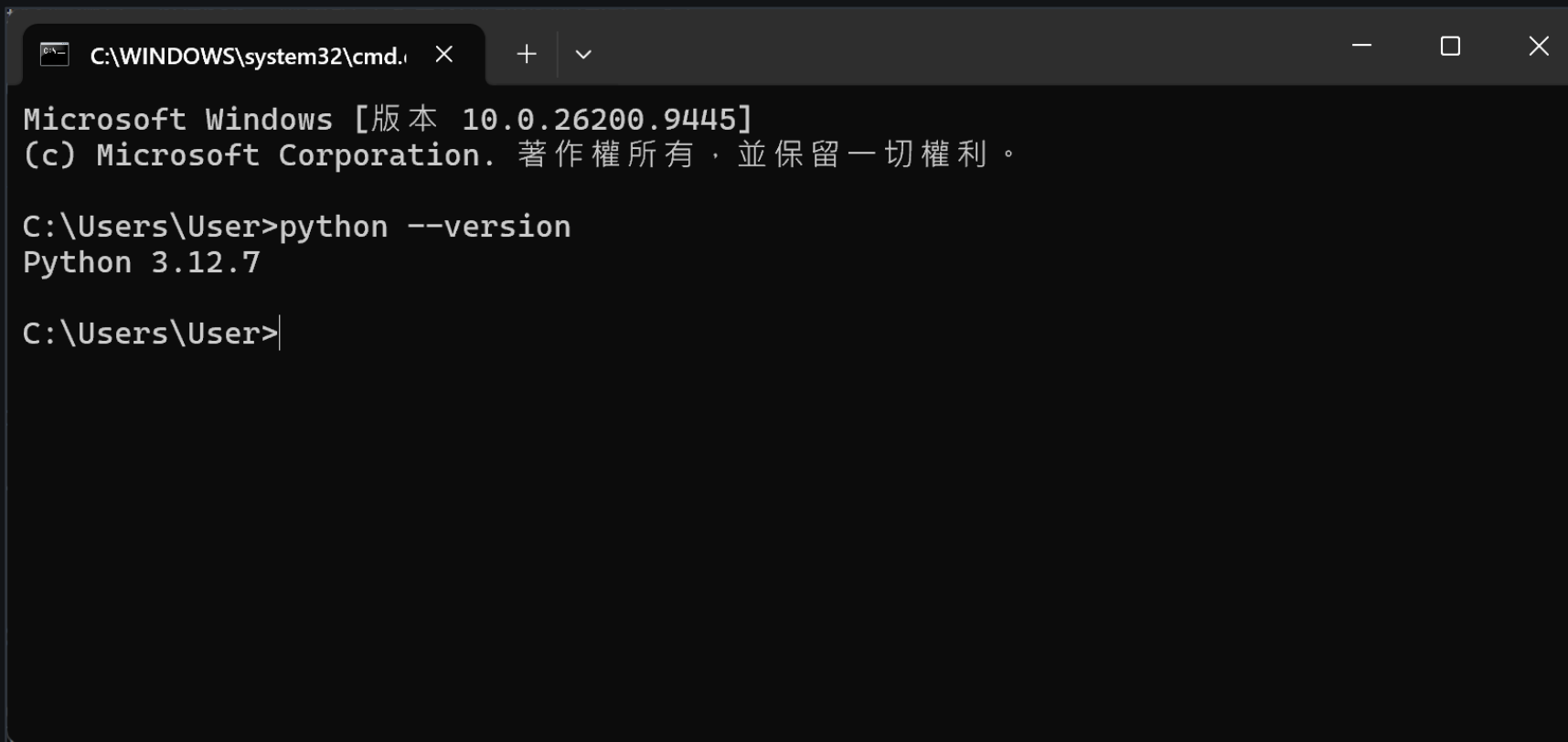


安裝路徑可自選（C 或 D 槽皆可）；但「Add python.exe to PATH」這個勾一定要打，否則終端機打 python 會失敗。

步驟 2

確認 Python 安裝成功

按 **Win + R** 輸入 `cmd`，再輸入 `python --version`



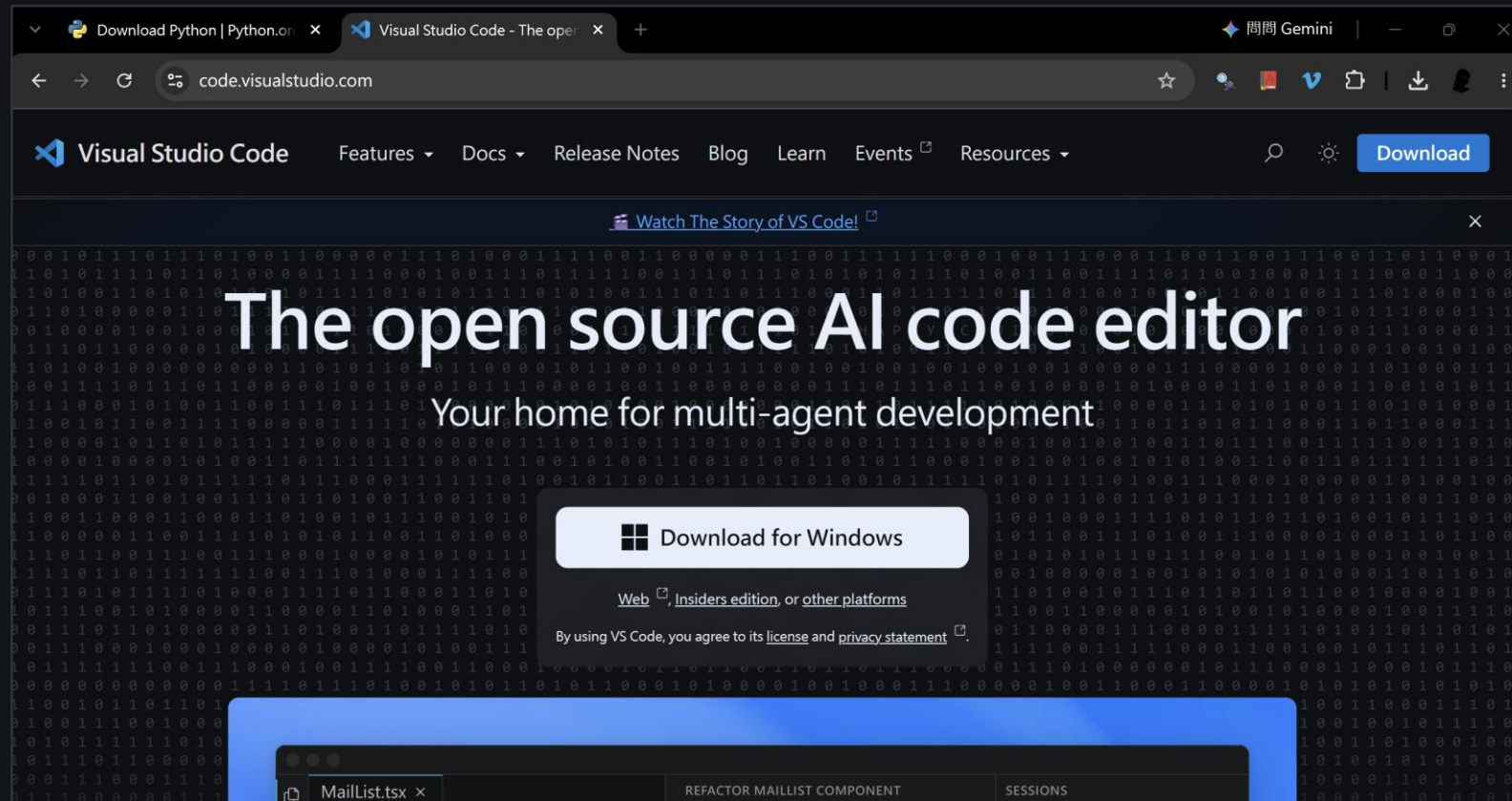
```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [版本 10.0.26200.9445]
(c) Microsoft Corporation. 著作權所有，並保留一切權利。

C:\Users\User>python --version
Python 3.12.7

C:\Users\User>
```

下載且安裝 VS Code

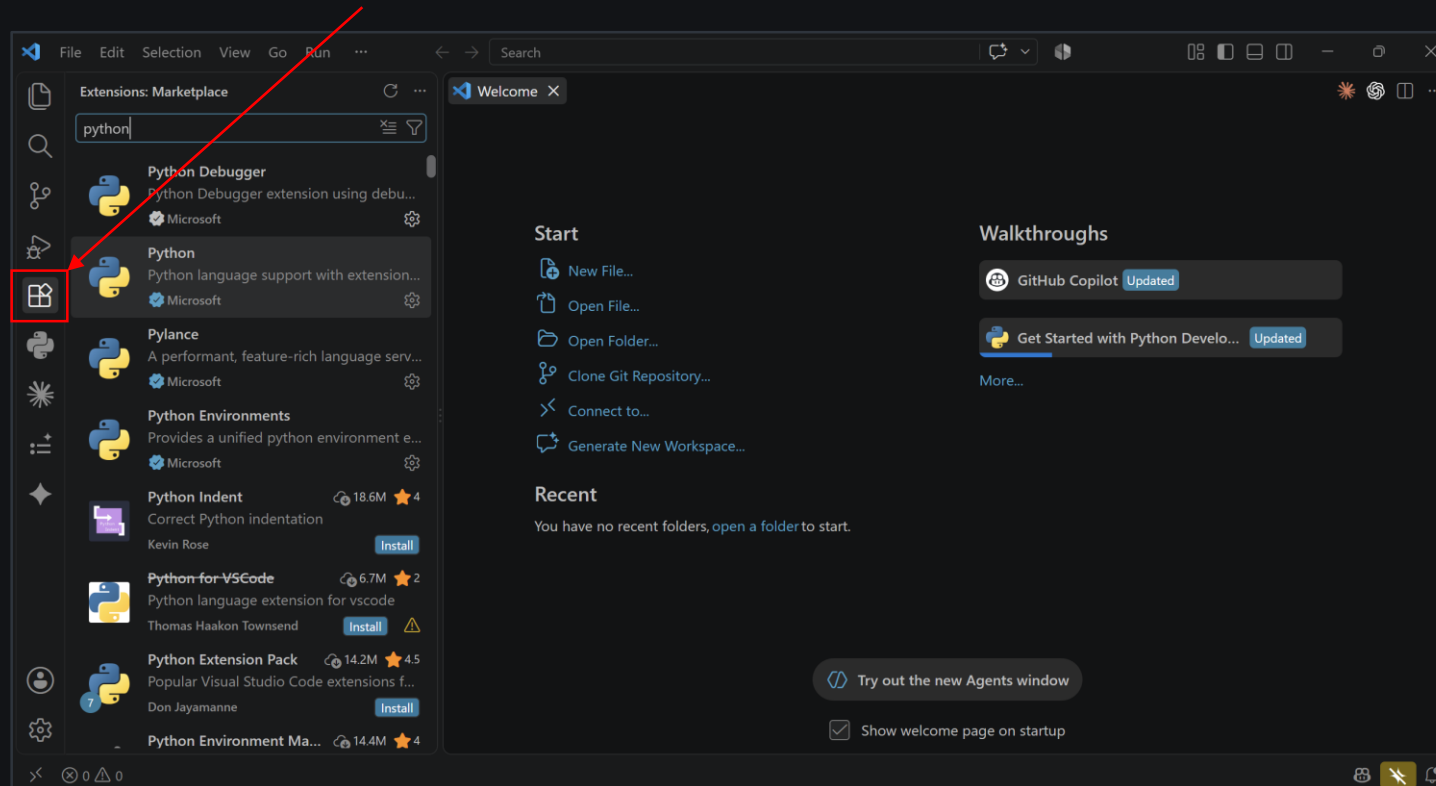
官網：<https://code.visualstudio.com/>



步驟 4

安裝 VS Code 的 Python 擴充套件

左側 Extensions → 搜尋 Python → 裝發行者 Microsoft 那個

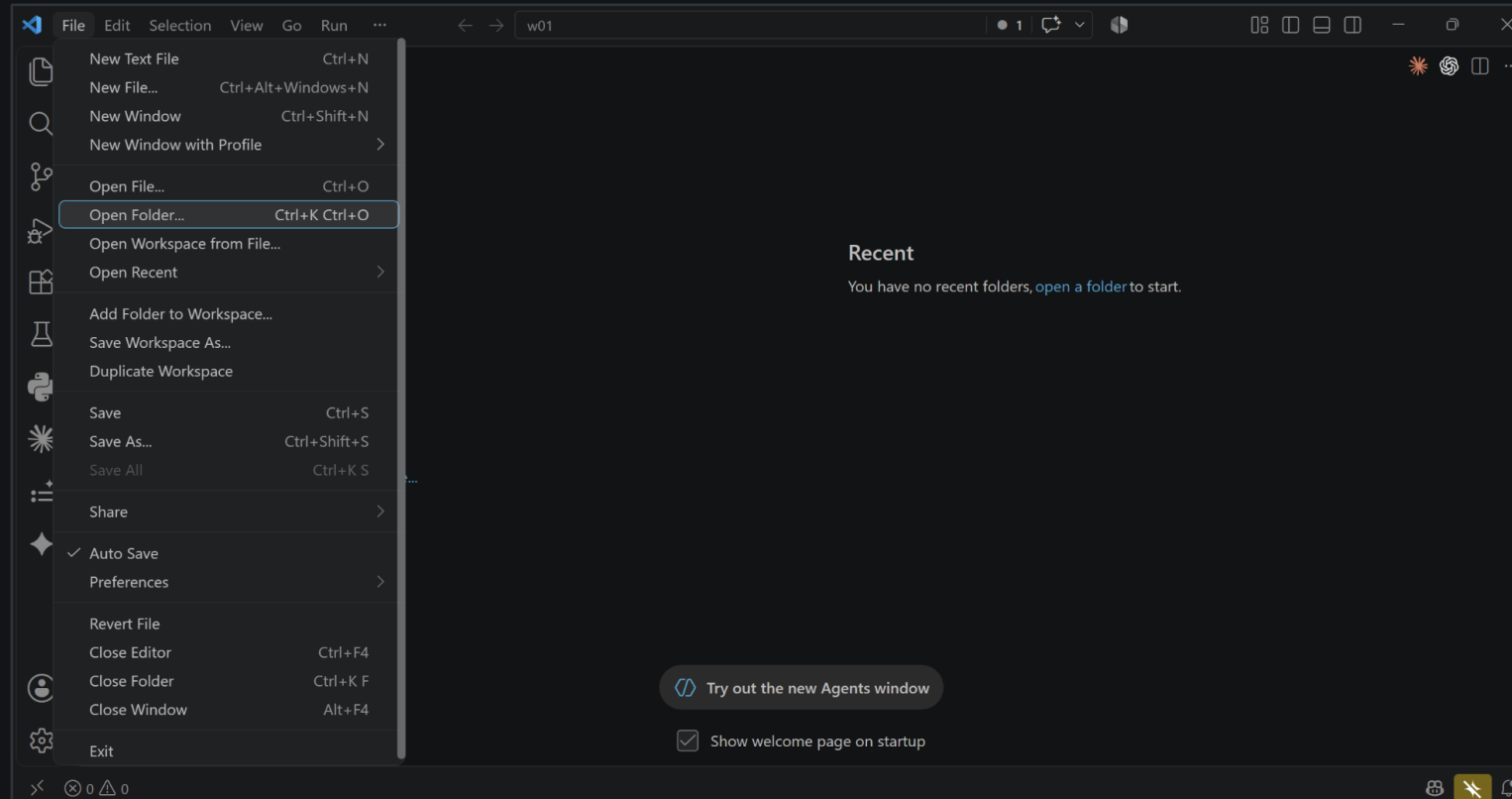


安裝 Microsoft 發行的 Python 擴充套件，會自動連帶安裝 Pylance 與 Debugger。

步驟 5

建立且開啟專案資料夾

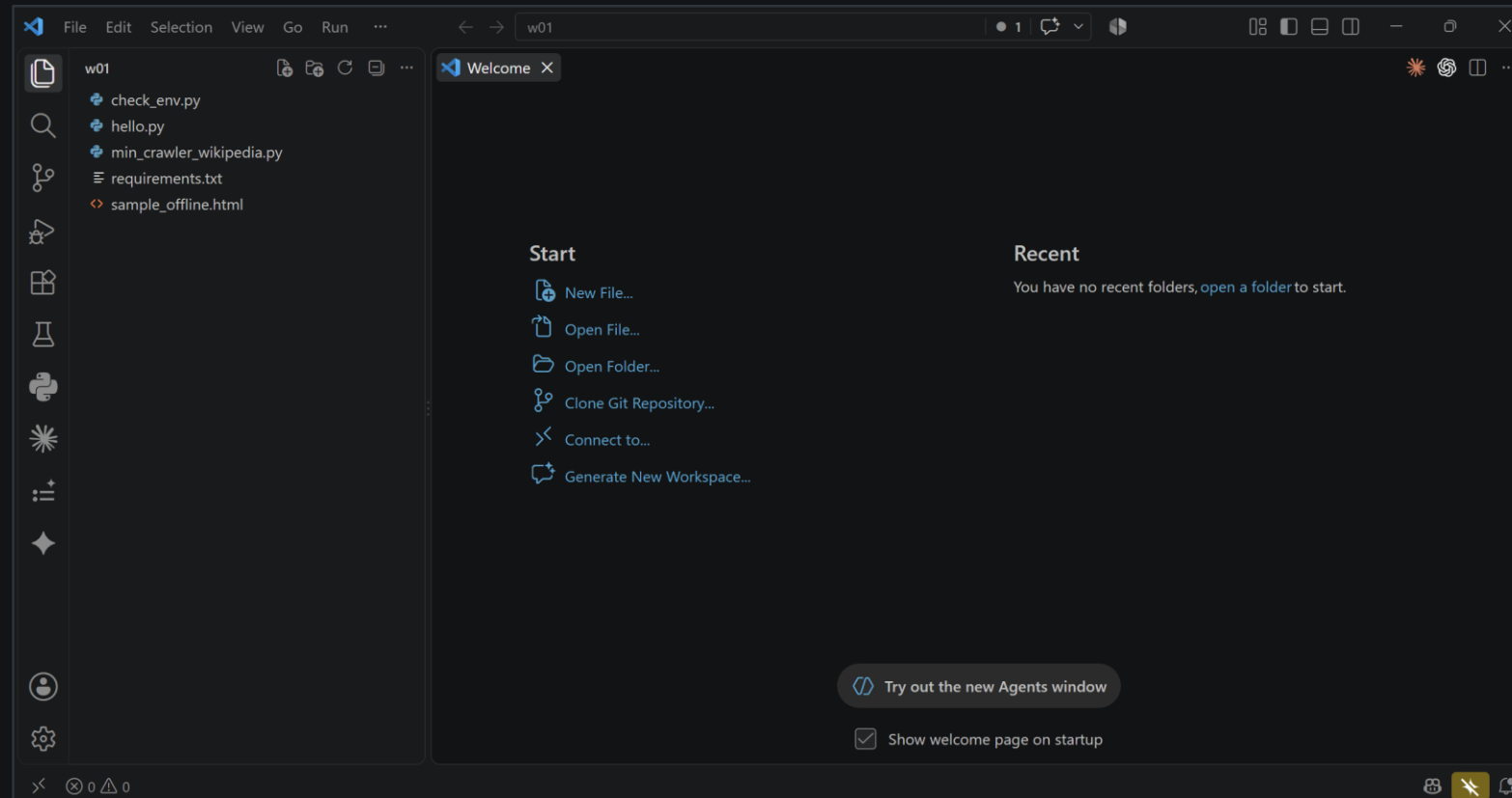
本機端建立路徑資料夾 → File ▶ Open Folder 開啟



步驟 6

建立/放置 Python 執行檔案

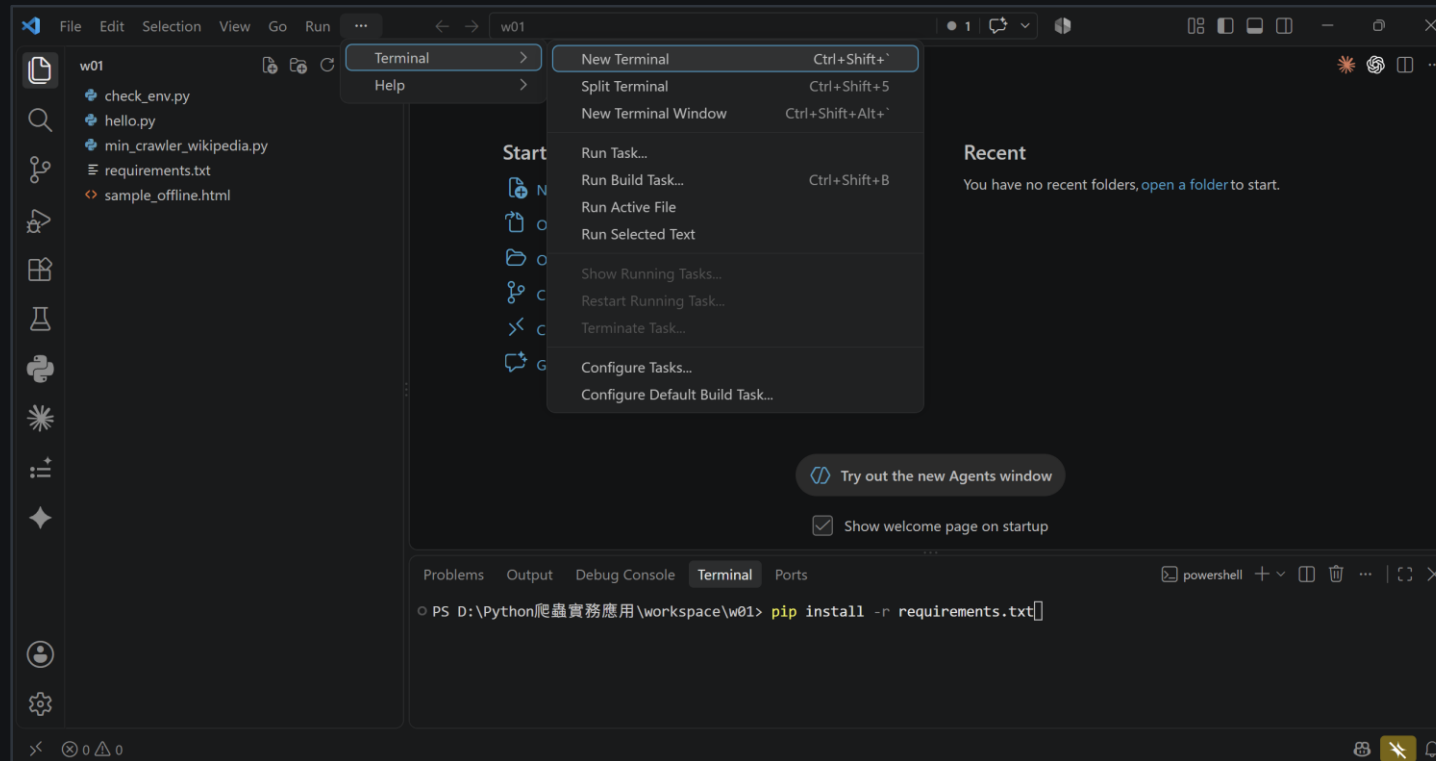
可從 VS Code 新增檔案撰寫 Python 程式碼，也可放置 Python 程式碼檔案執行



步驟 7

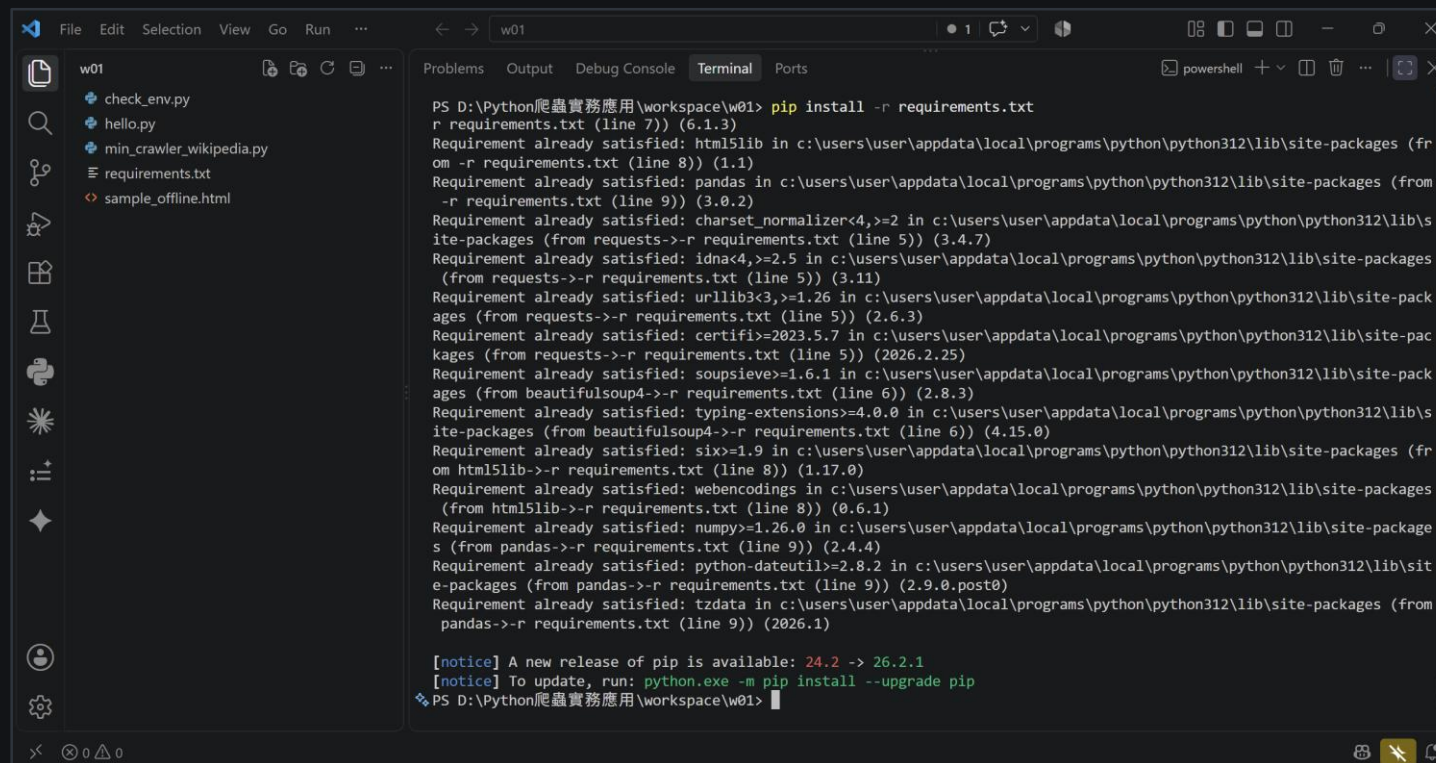
安裝需要的套件

開 New Terminal → `pip install -r requirements.txt`



看懂安裝結果

每行 Requirement already satisfied = 該套件 (與相依套件) 已裝好



```
PS D:\Python爬蟲實務應用\workspace\w01> pip install -r requirements.txt
r requirements.txt (line 7)) (6.1.3)
Requirement already satisfied: html5lib in c:\users\user\appdata\local\programs\python\python312\lib\site-packages (from
om -r requirements.txt (line 8)) (1.1)
Requirement already satisfied: pandas in c:\users\user\appdata\local\programs\python\python312\lib\site-packages (from
-r requirements.txt (line 9)) (3.0.2)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\user\appdata\local\programs\python\python312\lib\s
ite-packages (from requests->-r requirements.txt (line 5)) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in c:\users\user\appdata\local\programs\python\python312\lib\site-packages
(from requests->-r requirements.txt (line 5)) (3.11)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\user\appdata\local\programs\python\python312\lib\site-pack
ages (from requests->-r requirements.txt (line 5)) (2.6.3)
Requirement already satisfied: certifi=2023.5.7 in c:\users\user\appdata\local\programs\python\python312\lib\site-pac
kages (from requests->-r requirements.txt (line 5)) (2026.2.25)
Requirement already satisfied: soupsieve=1.6.1 in c:\users\user\appdata\local\programs\python\python312\lib\site-pack
ages (from BeautifulSoup4->-r requirements.txt (line 6)) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\users\user\appdata\local\programs\python\python312\lib\s
ite-packages (from BeautifulSoup4->-r requirements.txt (line 6)) (4.15.0)
Requirement already satisfied: six>=1.9 in c:\users\user\appdata\local\programs\python\python312\lib\site-packages (fr
om html5lib->-r requirements.txt (line 8)) (1.17.0)
Requirement already satisfied: webencodings in c:\users\user\appdata\local\programs\python\python312\lib\site-packages
(from html5lib->-r requirements.txt (line 8)) (0.6.1)
Requirement already satisfied: numpy>=1.26.0 in c:\users\user\appdata\local\programs\python\python312\lib\site-packag
es (from pandas->-r requirements.txt (line 9)) (2.4.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\user\appdata\local\programs\python\python312\lib\sit
e-packages (from pandas->-r requirements.txt (line 9)) (2.9.0.post0)
Requirement already satisfied: tzdata in c:\users\user\appdata\local\programs\python\python312\lib\site-packages (from
pandas->-r requirements.txt (line 9)) (2026.1)

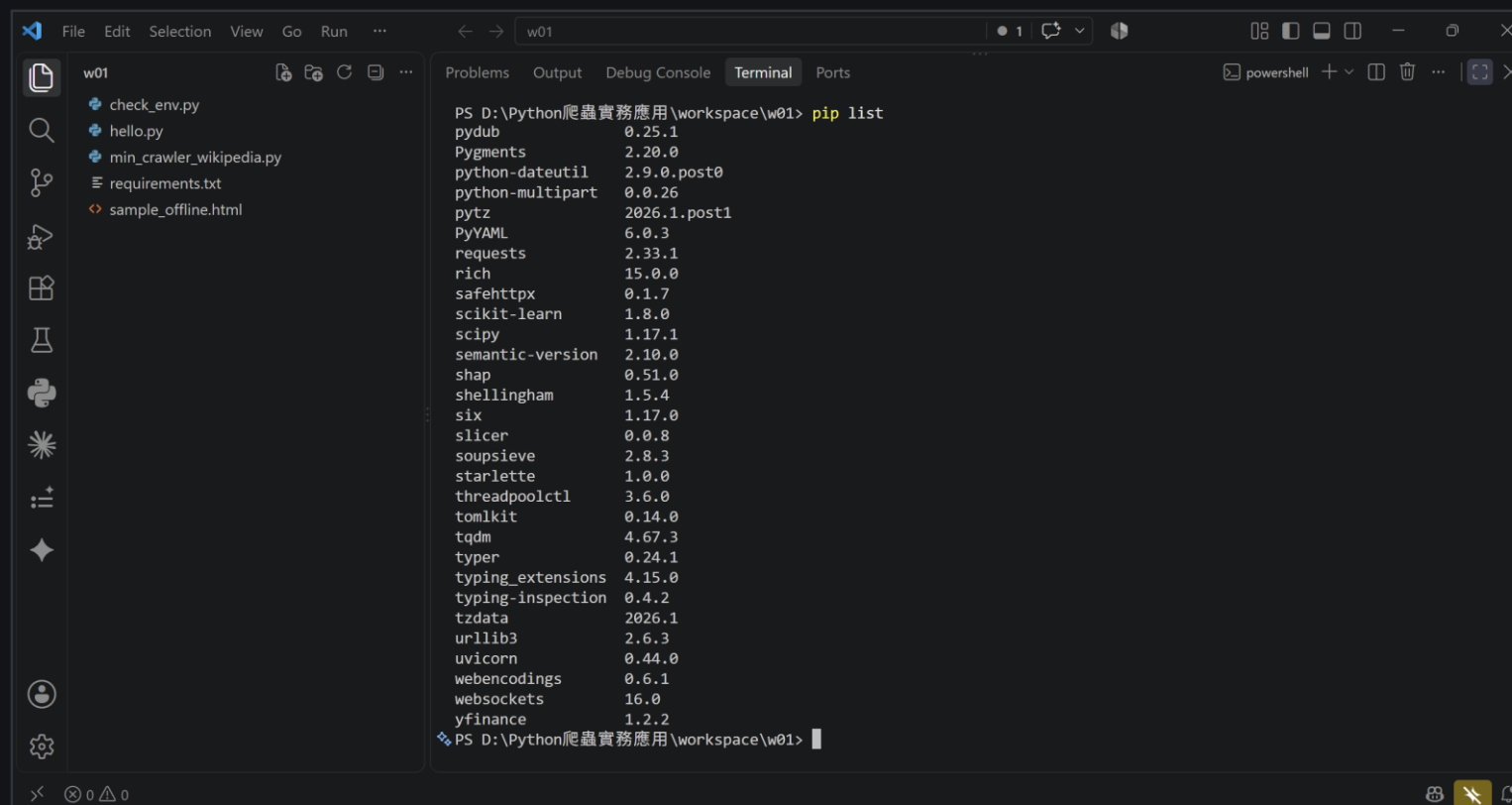
[notice] A new release of pip is available: 24.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS D:\Python爬蟲實務應用\workspace\w01>
```

若最後出現 pip 新版提示 (notice) , 那是提醒、不是錯誤。

步驟 9

檢視已安裝的套件

輸入 `pip list`，可看到所有套件與版本號



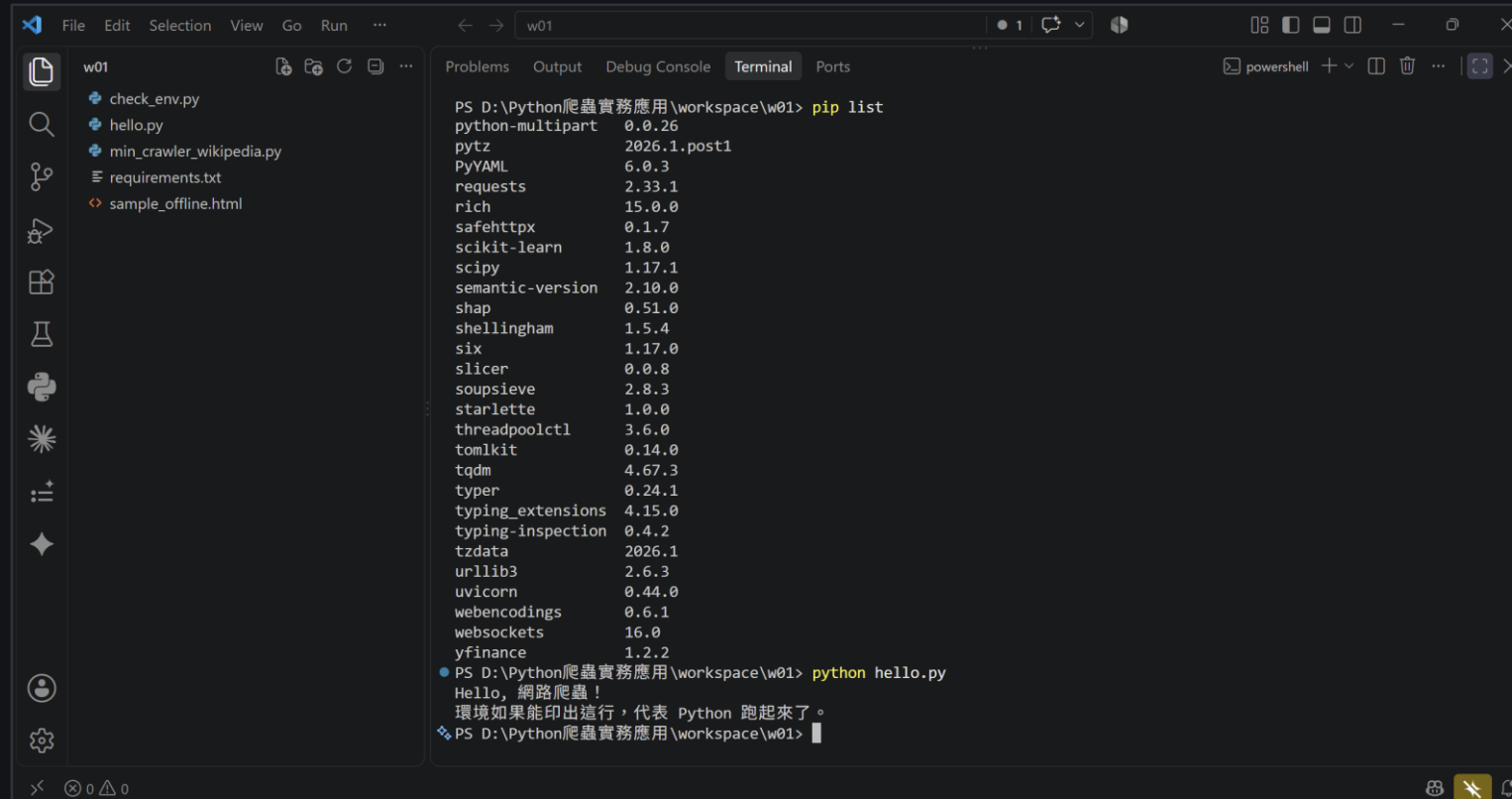
The screenshot shows the Visual Studio Code interface with a file explorer on the left and a terminal window on the right. The file explorer shows a workspace named 'w01' containing files: 'check_env.py', 'hello.py', 'min_crawler_wikipedia.py', 'requirements.txt', and 'sample_offline.html'. The terminal window is titled 'w01' and shows the command 'pip list' being executed in a PowerShell prompt. The output lists various installed Python packages and their versions.

```
PS D:\Python爬蟲實務應用\workspace\w01> pip list
pydub 0.25.1
Pygments 2.20.0
python-dateutil 2.9.0.post0
python-multipart 0.0.26
pytz 2026.1.post1
PyYAML 6.0.3
requests 2.33.1
rich 15.0.0
safehttpx 0.1.7
scikit-learn 1.8.0
scipy 1.17.1
semantic-version 2.10.0
shap 0.51.0
shellingham 1.5.4
six 1.17.0
slicer 0.0.8
soupsieve 2.8.3
starlette 1.0.0
threadpoolctl 3.6.0
tomlkit 0.14.0
tqdm 4.67.3
typer 0.24.1
typing_extensions 4.15.0
typing-inspection 0.4.2
tzdata 2026.1
urllib3 2.6.3
uvicorn 0.44.0
webencodings 0.6.1
websockets 16.0
yfinance 1.2.2
PS D:\Python爬蟲實務應用\workspace\w01>
```

步驟 10

執行 hello.py

輸入 `python hello.py`，看到問候文字就代表能跑

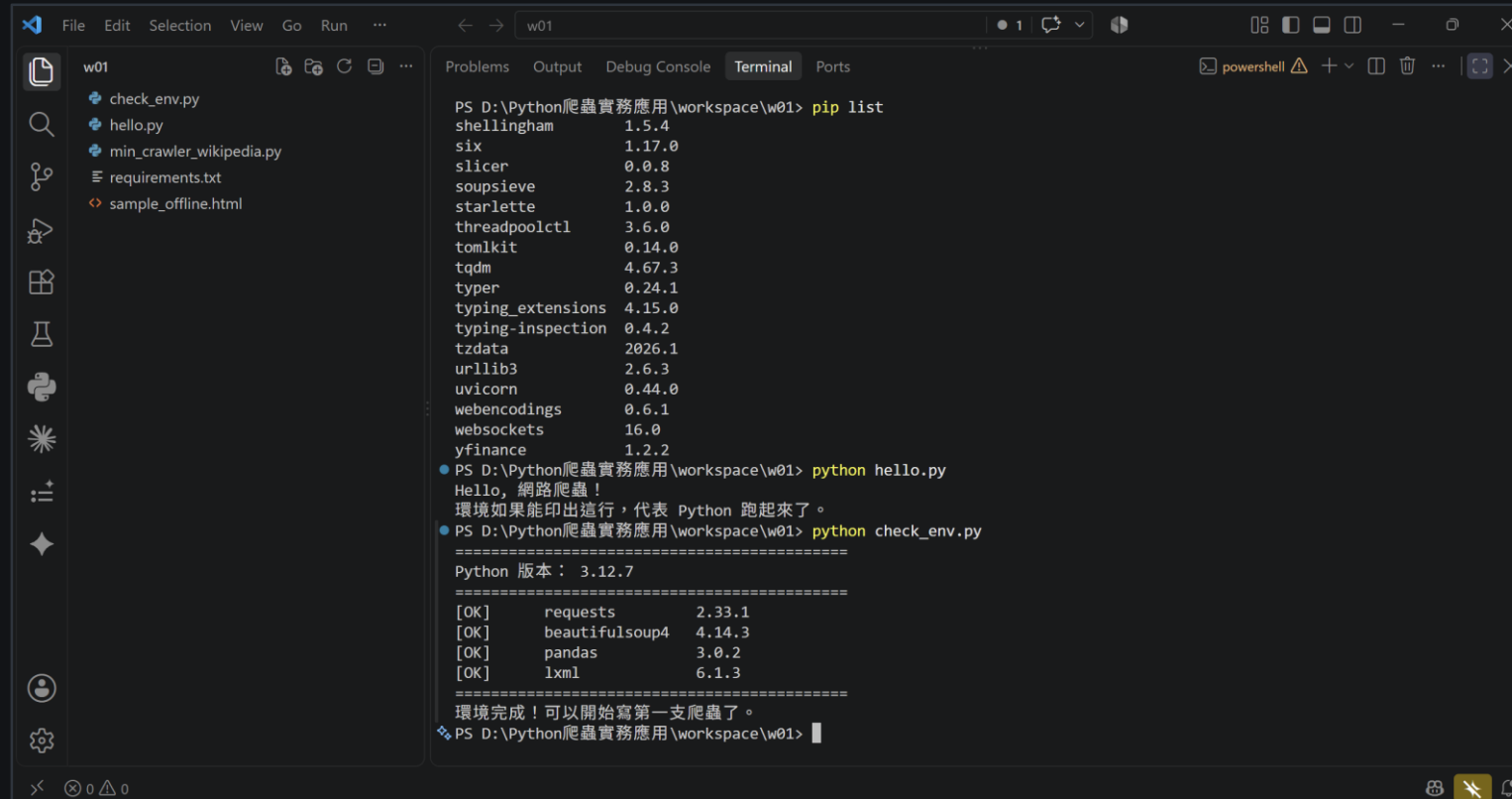


```
PS D:\Python爬蟲實務應用\workspace\w01> pip list
python-multipart 0.0.26
pytz 2026.1.post1
PyYAML 6.0.3
requests 2.33.1
rich 15.0.0
safehttpx 0.1.7
scikit-learn 1.8.0
scipy 1.17.1
semantic-version 2.10.0
shap 0.51.0
shellingham 1.5.4
six 1.17.0
slicer 0.0.8
soupsieve 2.8.3
starlette 1.0.0
threadpoolctl 3.6.0
tomlkit 0.14.0
tqdm 4.67.3
typer 0.24.1
typing_extensions 4.15.0
typing-inspection 0.4.2
tzdata 2026.1
urllib3 2.6.3
uvicorn 0.44.0
webencodings 0.6.1
websockets 16.0
yfinance 1.2.2
PS D:\Python爬蟲實務應用\workspace\w01> python hello.py
Hello, 網路爬蟲!
環境如果能印出這行，代表 Python 跑起來了。
PS D:\Python爬蟲實務應用\workspace\w01>
```

步驟 11

執行 check_env.py 驗證環境

輸入 `python check_env.py` → 四個 [OK] 代表環境完成



The screenshot shows a Visual Studio Code window with a terminal open. The terminal output is as follows:

```
PS D:\Python爬蟲實務應用\workspace\w01> pip list
shellingham 1.5.4
six 1.17.0
slicer 0.0.8
soupsieve 2.8.3
starlette 1.0.0
threadpoolctl 3.6.0
tomlkit 0.14.0
tqdm 4.67.3
typer 0.24.1
typing_extensions 4.15.0
typing-inspection 0.4.2
tzdata 2026.1
urllib3 2.6.3
uvicorn 0.44.0
webencodings 0.6.1
websockets 16.0
yfinance 1.2.2

PS D:\Python爬蟲實務應用\workspace\w01> python hello.py
Hello, 網路爬蟲!
環境如果能印出這行，代表 Python 跑起來了。

PS D:\Python爬蟲實務應用\workspace\w01> python check_env.py
=====
Python 版本： 3.12.7
=====
[OK] requests 2.33.1
[OK] beautifulsoup4 4.14.3
[OK] pandas 3.0.2
[OK] lxml 6.1.3
=====
環境完成！可以開始寫第一支爬蟲了。
PS D:\Python爬蟲實務應用\workspace\w01>
```

完成確認

環境完成檢查清單

- ✓ `python --version` 有印出版本號
- ✓ VS Code 已裝 Microsoft 的 Python 擴充
- ✓ `pip list` 看得到 `requests` / `pandas` 等套件
- ✓ `python hello.py` 有輸出
- ✓ `python check_env.py` 四個都 [OK]